Host Cytokine and Chemokine Profiles Linked to Survival in Mice Challenged with *Burkholderia pseudomallei*

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BACKGROUND

Burkholderia pseudomallei is a gram-negative bacteria environmental bacteria that can cause melioidosis. It is classified as a Tier 1 select agent and a biosafety level 3 pathogen. Currently, there are no licensed vaccines for *B. pseudomallei*.

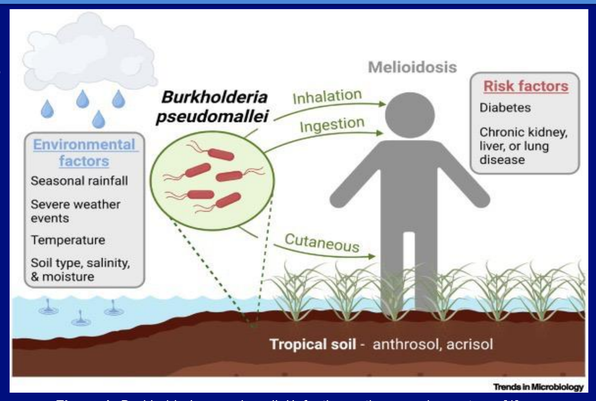

Figure 1. *Burkholderia pseudomallei* infection pathway and symptoms [1].

METHODS

For both experiments, C57BL/6 mice were challenged with *B. pseudomallei* by a whole-body aerosol route. Within the experimental timelines, blood samples were collected several timepoints: pre-vaccination, booster vaccination, multiple post vaccination and post challenge time points.

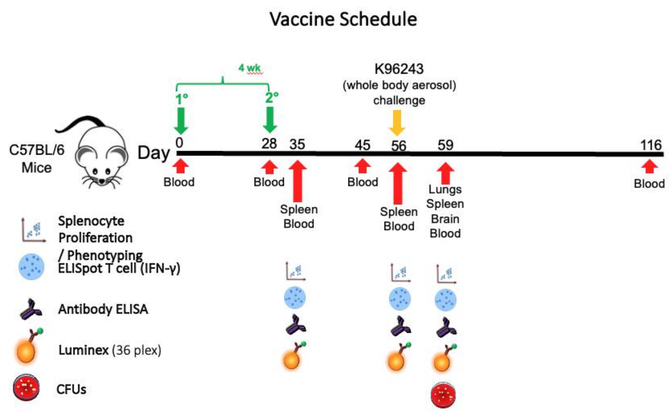

Figure 2. Challenge Experiment Design. Timeline that includes treatment, challenge and data extracted.

Phase 1	Phase 2	
UNR + OMV: Hcp1-TL + CPS:CRM197 + Alu + CpG	**DNA (Hcp1)**	**DNA (Hcp1) A/C:** DNA (Hcp1) + alhydrogel + CPG
Self Amplifying RNA: Hcp1-SAM:LNP (L014) + CPS:CRM197 + Alu + CpG	**DNA (Hcp1 - secreted)**	**DNA (Hcp1-secreted) A/C:** DNA (Hcp1-secreted) + alhydrogel + CPG
Conventional RNA: Hcp1-SAM:LNP (L018) + CPS:CRM197 + Alu + CpG	**CPS-CRM197:** CPS-CRM197 + alhydrogel + CPG	**DNA (Hcp1) + CPS-CRM197:** DNA (Hcp1); DNA (Hcp1-secreted) and CPS-CRM197 + alhydrogel + CPG
DNA Vaccine: pWRG/Hcp1(v2) DNA + CPS:CRM197 + Alu + CpG	**Hcp1 + CPS-CRM197:** Hcp1 + CPS-CRM197 + alhydrogel + CPG	**DNA (Hcp1-secreted) + CPS-CRM197:** DNA (Hcp1-secreted) and CPS-CRM197+ alhydrogel + CPG
UNR Positive Control		

Differences within Phase 1 and Phase 2

Within Phase 1, the mice were challenged with high bacteria dose (8 LD_{50}). This high dose led to low survival rates (≤40%). In Phase 2, the challenge dose of *B. pseudomallei* was lowered (5 LD_{50}) with the goal of increasing the range of survival rates within the mice.

RESULTS

Survival Rate Gap Between Experiments

As mentioned, the survival rates within Phase 1 maintained a small range; while Phase 2 had a larger range of survival rates. The wider range of survival rates allows for a more comprehensive analysis of drivers behind survival.

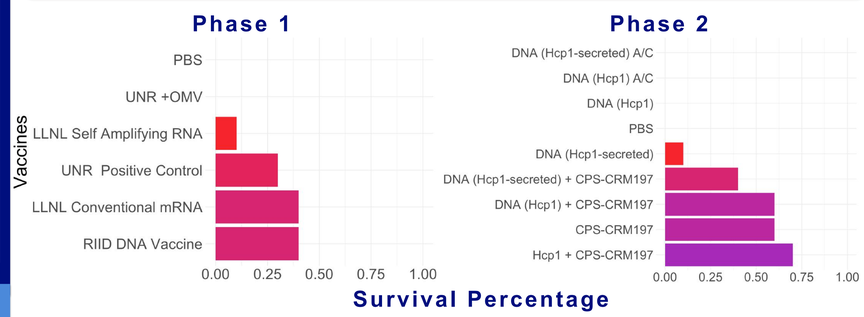

Figure 3. Survival Percentages by vaccine platform for both Phase 1 and Phase 2.

Survival Curves

Time-to-death (TTD) data presented in these graphs for Phase 1 and 2 provide insight into disease progression and vaccine efficacy.

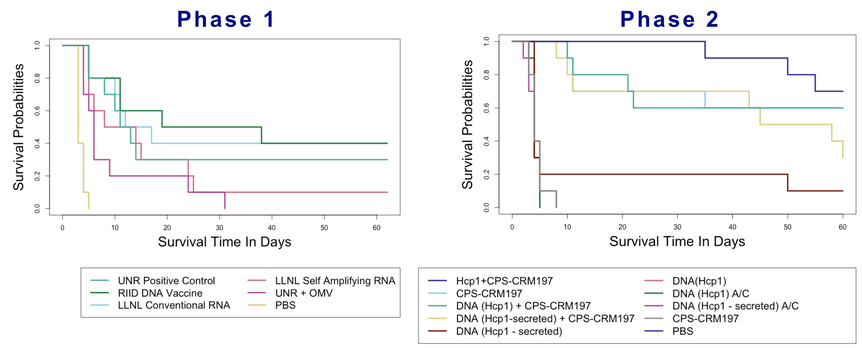

Figure 4. Survival Curves for Phases 1 and 2 of the challenge studies.

Antibody Titer Tracking

For analyzing the antibody titers collected from the study, the titers were mapped to each vaccine group. Tracking the antibody titers revealed a trend that the higher the antibody response was post-vaccination the lower the response would be post-challenge. Meaning, a higher antibody response to the vaccine indicates a higher probable survival rate post challenge.

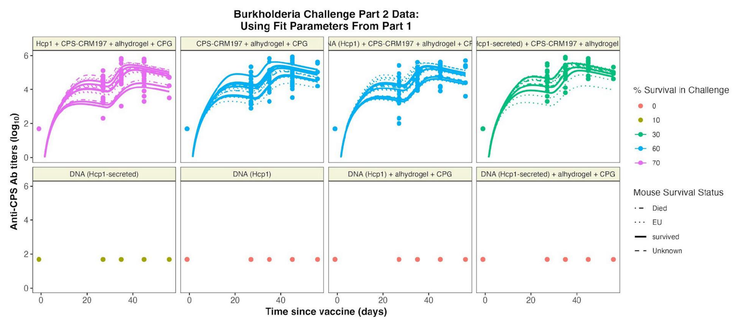

Figure 5. Antibody titer tracking for Phase 2. Mapped with parameters from Phase 1.

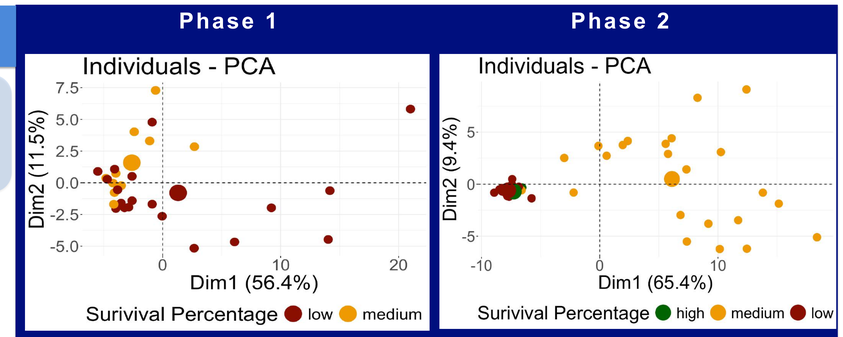

Figure 6. PCA models of Phase 1 and Phase 2 data post challenge.

CONCLUSIONS

Random Forest Analysis

To analyze any potential relation between survival rates and the chemokine/cytokine responses, random forest models were implemented. The survival rates into three bins: low (≤30%), medium (30% ≤ 60%), high (>60%). All data is included in the confusion matrix, but the training model and the validation model both maintained an error rate of ~10%.

Actual \ Prediction	Low	Med	High
Low	23	0	2
Med	0	15	0
High	0	2	3

Figure 7. Confusion matrix generated by random forest model for Phase 2 post-challenge data.

Important Chemokines and Cytokines

Within the random forest model, chemokines and cytokines from the harvested lung tissue have the highest correlation to survival predictions.

Cytokine/ Chemokine	Function
IFN-γ	Stimulates macrophages.
IL15	Stimulates proliferation and activation of NK cells and CD8+ T cells.
CCL7	Stimulated inflammation by attracting macrophages and monocytes.
GM-CSF	Granulocyte-macrophage colony stimulating factor.
CXCL10	Chemoattractant. CXCR3 receptor, contributes to neuronal and glial cell death.
IL17A	Pro-inflammatory and can contribute to oxidative stress.
IL18	Pro-inflammatory. Involved in apoptosis and pyroptosis.
IL4	Contributes to differentiation of helper T cells.
CCL2	Pro-inflammatory, chemoattractant for monocytes and macrophages.
IL13	Chemoattractant.

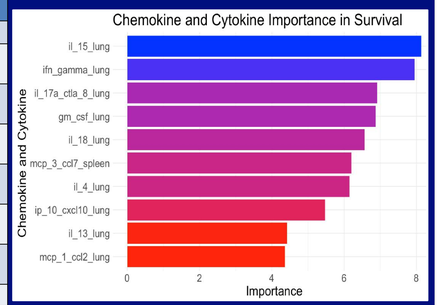

Figure 8. Feature importance for classification in random forest model.

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